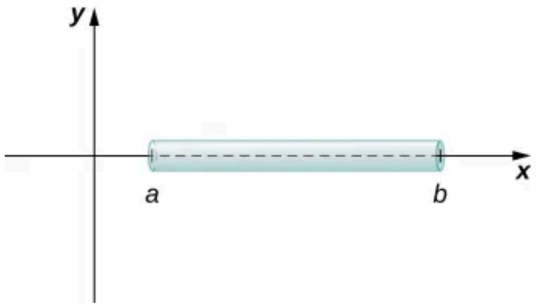
1 Mass of a thin rod

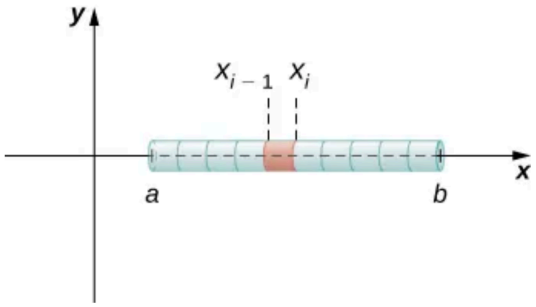
We can use integration to develop a formula for calculating mass based on a density function. First we consider a thin rod or wire. Orient the rod so it aligns with the x-axis, with the left end of the rod at $x = a$ and the right end of the rod at $x = b$. If the rod has constant linear density ρ, given in



terms of mass per unit length, then the mass M of the rod is just

$$M = \text{density} \times \text{length} = \rho\,(b-a). \tag{1}$$

Now suppose the density of the rod varies from point to point along its length according to a continuous linear *density function* $\rho\,(x)$. To calculate the mass, partition the rod into n subintervals of equal length $\Delta x = \frac{b-a}{n}$.



If x_i^* is an arbitrary point in the ith subinterval $[x_{i-1}, x_i]$, then since ρ is continuous it does not change much over the ith subinterval as long as its length is small (which will be the case if n is large), so the density is approximately the constant value $\rho\,(x_i^*)$ over the ith subinterval, and the mass M_i of the ith subinterval is approximately given by Equation (1):

$$M_i \approx \rho\,(x_i^*)\Delta x.$$

Adding the masses of all the segments gives us an approximation for the mass of the entire rod:

$$M = \sum_{i=1}^{n} M_i \approx \sum_{i=1}^{n} \rho\,(x_i^*)\Delta x,$$

which is a Riemann sum. Since taking n larger gives a better approximation, we define the exact mass by taking the limit as $n \to \infty$ (which exists since ρ is continuous):

$$\boxed{M := \lim_{n\to\infty} \sum_{i=1}^{n} \rho\,(x_i^*)\Delta x = \int_a^b \rho\,(x)dx.}$$

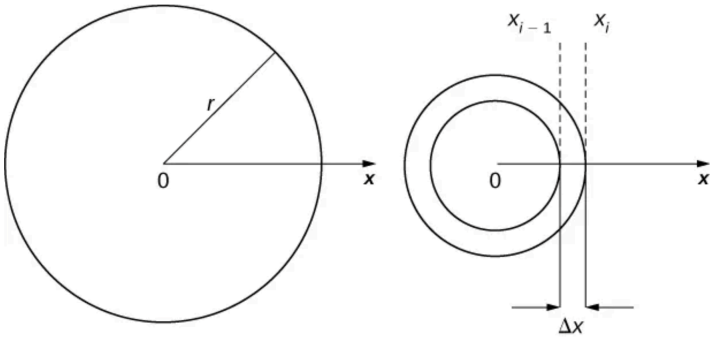
1. Consider a thin rod oriented along the x-axis over the interval $[\frac{\pi}{2}, \pi]$. If the density of the rod is given by $\rho(x) = \sin(x)$, what is the mass of the rod?

2. Consider a thin rod oriented along the x-axis over the interval $[1, 3]$. If the density of the rod is given by $\rho(x) = 2x^2 + 3$, what is the mass of the rod?

2 Mass of a thin disk

Now consider a thin disk of radius R whose center is at the origin in the xy-plane. Assume the density is given by a continuous *radial density function* $\rho(r)$, which varies only along the disk's radius (that is, the mass is distributed in a rotationally-symmetric way about the origin).

Similar to what we did before, partition the interval $[0,R]$ into n equally spaced subintervals $[r_{i-1}, r_i]$ of width $\Delta r = \frac{R}{n}$. This partitions the disk into n concentric annular regions of width Δr. The inner radius of the ith annular region is r_{i-1} and the outer radius is r_i.



The area of the ith annular region is

$$\begin{aligned} A_i &= \pi r_i^2 - \pi r_{i-1}^2 \\ &= \pi(r_i^2 - r_{i-1}^2) \\ &= \pi(r_i + r_{i-1})(r_i - r_{i-1}) \\ &= \pi(r_i + r_{i-1})\Delta r \\ &= 2\pi \frac{r_i + r_{i-1}}{2}\Delta r. \end{aligned}$$

Using $r_i^* \approx \frac{r_i + r_{i-1}}{2}$ to approximate the average radius of the ith annular region, we have

$$A_i \approx 2\pi r_i^* \Delta r.$$

Since ρ is continuous, if Δr is small then the density is approximately the constant value $\rho(r_i^*)$ over the ith annular region, so the mass of this region is approximately

$$M_i \approx 2\pi r_i^* \rho(r_i^*)\Delta r.$$

Adding up the masses of the annular regions, the mass M of the entire disk is approximately,

$$M = \sum_{i=1}^{n} M_i \approx \sum_{i=1}^{n} 2\pi r_i^* \rho(r_i^*)\Delta r.$$

Recognizing this as a Riemann sum, we take the limit as $n \to \infty$ (which exists since ρ was assumed to be continuous), and define the mass of the thin disk by

$$\boxed{M := \lim_{n\to\infty} \sum_{i=1}^{n} 2\pi r_i^* \rho(r_i^*)\Delta r = \int_0^R 2\pi r \rho(r)\, dr}.$$

3. Calculate the mass of a thin disk of radius 4 whose radial density function is given by $\rho(r) = \sqrt{r}$.

4. Let $\rho(r) = 3r + 2$ represent the radial density of a thin disk of radius 2. Calculate the mass of the disk.